

Problem Definition

- Need for timely warnings of weather and tsunami events
- Difficulty in integrating various sensor data
- Challenges in effective communication of alerts

System Context

The Smart Monitoring System for Weather and Tsunami Detection (SMSWTD) is designed to work in a dynamic and challenging environment. Its main goal is to deliver timely, useful insights during disasters by combining advanced sensors, data analytics, and clear communication channels. SMSWTD adapts to changing conditions like severe weather and oceanic disturbances, using tools such as satellites, aircraft, underwater sensors, and machine learning to create a complete picture of potential risks. It also connects emergency responders, government agencies, researchers, and the public to ensure everyone gets the information they need when it matters most.

Operating Environment

SMSWTD is designed to function in diverse and often extreme conditions. Its operating environment includes the following:

SMSWTD must be capable of working in diverse geographies including cities, coastal regions, remote areas, and mountains. Components of SMSWTD are also present in deep oceans in the form of underwater sensors. In the atmosphere, storm-prone areas will be challenging for satellite and airplane surveillance. Similarly, severe weather conditions, such as thunderstorms, typhoons, hurricanes and oceanic disruptions, can affect the working of SMSWTD. Likewise, seismic activity or earthquakes will have an impact on tsunami forecasts.

Limitations of infrastructure in remote areas with inadequate networks for communication must also be taken into account. During disasters, there can be power outages and damaged facilities. Moreover, there can be places at risk with poor infrastructure for disaster preparedness.

Other environmental factors, such as rusting of underwater sensors in saltwater, harsh atmospheric conditions that interfere with signals, and conditions of high pressure for oceanic sensors are also challenging for this system.

User interactions (Stakeholders)

The system is made to meet the needs of a wide range of users, each with specific requirements. It is used by government organizations for policy planning and disaster response. The government also requires real-time risk assessments and complete technical data. The disaster responders use the system for rescue operations, who need clear actionable alerts to make decisions quickly. The public or civilians relies on the system and expects clear alerts via SMS, apps, or other media for personal safety and evacuation.

System boundary

The SMSWTD's scope is defined by the boundaries within which it functions. Satellite imaging, aircraft-based sensors, and underwater seismic and pressure sensors are some of its key data sources. Data collection, risk assessment, and alert distribution are its main responsibilities. However, it is not responsible for the post-disaster infrastructure restoration or physical disaster management such as evacuation or direct involvement. Furthermore, the system might not function in areas with inadequate infrastructure or in dangerous settings like conflict zones.